GATE ME 90 Days Challenge

Day 69/90 · Practice Set with Detailed Solutions

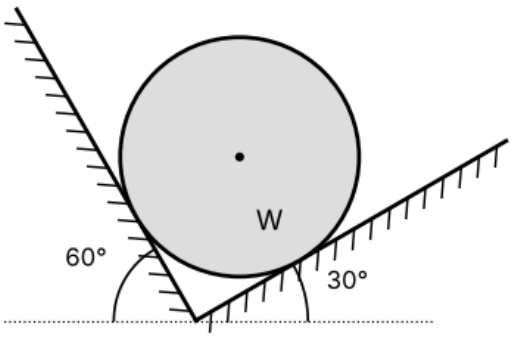
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mechanics + Strength of Materials	**EM & SOM Revision**	**31 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a full revision of Engineering Mechanics (equilibrium of a sphere in a groove, trusses, ladder friction, projectiles, power and impact) and Strength of Materials (elastic constants, tapering bars, SFD and BMD, section modulus, cantilever deflection, torsion of fixed shafts, thin cylinders with joint efficiency, effective length of columns and stress transformation). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

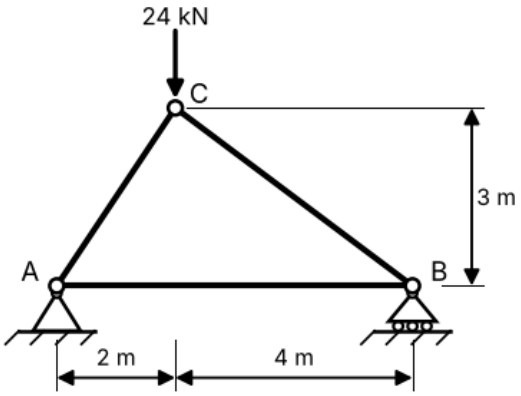
A smooth sphere of weight W = 100 N rests in a V-groove whose faces make 60° and 30° with the horizontal, as shown. The reactions on the sphere from the 30° face and the 60° face are, respectively,



(A) 50 N and 86.6 N
(B) 86.6 N and 50 N
(C) 100 N and 57.7 N
(D) 57.7 N and 115.5 N

Q2 NAT

The truss ABC shown is pinned at A and supported on a roller at B, and carries a vertical load of 24 kN at C. The force in member AB (in kN, round off to two decimal places) is _____ (tension positive)



Q3 MCQ

A uniform ladder rests on a rough horizontal floor ($\mu = 0.3$) and leans against a smooth vertical wall. The least angle the ladder can make with the floor without slipping is about

(A) 73.3°
(B) 30.9°
(C) 59.0°
(D) 45.0°

Q4 NAT

A ball is projected from level ground with a speed of 20 m/s. Taking $g = 9.81$ m/s², the smaller of the two angles of projection (in degrees) for which its horizontal range is 30 m (round off to two decimal places) is _____

Q5 MCQ

A pump lifts 2 m³ of water per minute through a height of 20 m. If its overall efficiency is 70 %, the power input to the pump is about

(A) 6.54 kW
(B) 9.34 kW
(C) 4.58 kW
(D) 13.08 kW

Q6 NAT

A 1 kg sphere moving at 4 m/s collides directly with a 2 kg sphere moving at 2 m/s in the opposite direction. The coefficient of restitution is 0.5. The loss of kinetic energy in the impact (in J) is _____

Q7 MCQ

For an isotropic material, Young's modulus is found to be 2.4 times its modulus of rigidity. The ratio of the bulk modulus to the modulus of rigidity (K/G) is

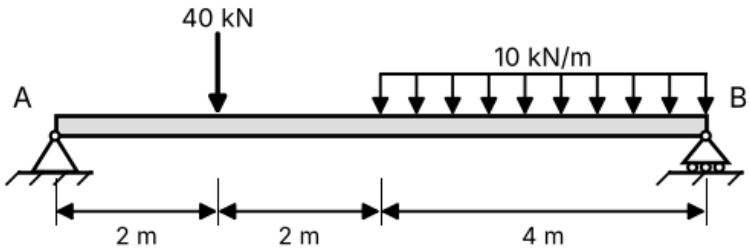
(A) 4/3
(B) 3/4
(C) 5/3
(D) 2

Q8 NAT

A steel bar 1 m long tapers uniformly from 40 mm diameter at one end to 20 mm diameter at the other. It carries an axial pull of 50 kN. Taking E = 200 GPa, its elongation (in mm, round off to three decimal places) is _____

Q9 NAT

The simply supported beam AB shown (span 8 m) carries a 40 kN point load at 2 m from A and a UDL of 10 kN/m over the 4 m next to B. The maximum bending moment in the beam (in kN·m) is _____



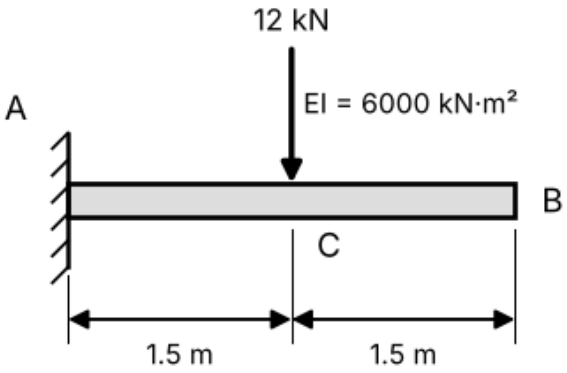
Q10 MCQ

A beam has a triangular cross-section of base b and height h, with the base horizontal. Its section modulus for the fibre at the top vertex (about the centroidal axis parallel to the base) is

(A) $bh^2/12$
(B) $bh^2/36$
(C) $bh^2/6$
(D) $bh^2/24$

Q11 NAT

The cantilever AB shown (length 3 m, EI = 6000 kN·m²) is fixed at A and carries a 12 kN load at its mid-length C. The deflection of the free end B (in mm, round off to two decimal places) is _____



Q12 MCQ

A shaft AB of length L is fixed at both ends and a torque T is applied at a section C, at a distance a from A and b from B (a + b = L). The fixing torque at end A is

(A) Ta/L
(B) Tb/L
(C) T/2
(D) Tab/L^2

Q13 MCQ

A thin cylindrical shell of 800 mm internal diameter and 8 mm wall thickness carries an internal pressure of 2 MPa. The longitudinal joint efficiency is 80 % and the circumferential joint efficiency is 50 %. The hoop stress (taking the joint into account) is

(A) 100 MPa
(B) 62.5 MPa
(C) 125 MPa
(D) 156.25 MPa

Q14 MCQ

The effective length of a column of length L with one end fixed and the other end hinged (pinned), as used in Euler's formula, is

(A) L/√2
(B) L/2
(C) 2L
(D) L

Q15 NAT

At a point in a body under plane stress, σx = 60 MPa, σy = 20 MPa and τxy = +15 MPa (positive: acting in the +y direction on the +x face). The normal stress on a plane whose normal is at 30° counter-clockwise from the x-axis (in MPa, round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(B) 86.6 N and 50 N
Q2	NAT	10.67 (accept 10.60 to 10.75)
Q3	MCQ	(C) 59.0°
Q4	NAT	23.69 (accept 23.50 to 23.85)
Q5	MCQ	(B) 9.34 kW
Q6	NAT	9
Q7	MCQ	(A) 4/3
Q8	NAT	0.398 (accept 0.395 to 0.401)
Q9	NAT	80
Q10	MCQ	(D) $bh^2/24$
Q11	NAT	5.63 (accept 5.58 to 5.67)
Q12	MCQ	(B) Tb/L
Q13	MCQ	(C) 125 MPa
Q14	MCQ	(A) $L/\sqrt{2}$
Q15	NAT	62.99 (accept 62.70 to 63.30)

Detailed Solutions

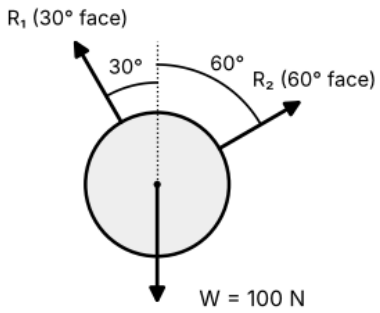
Q1 **Answer: (B) 86.6 N and 50 N**

Concept: A smooth surface pushes along its normal. The normal to a face inclined at α to the horizontal makes α with the vertical.

R_1 (30° face) acts at 30° from the vertical, R_2 (60° face) at 60° from the vertical, on opposite sides; the two normals are at 90° to each other.

Horizontal: $R_1 \sin 30° = R_2 \sin 60°$. Vertical: $R_1 \cos 30° + R_2 \cos 60° = 100$. Hence $R_1 = W \cos 30° =$ **86.6 N** and $R_2 = W \cos 60° =$ **50 N (B)**.

Exam tip: The flatter face carries the larger share of the weight; Lami's theorem gives the same result in one line.



Q2 **Answer: 10.67 (accept 10.60 to 10.75)**

Concept: Find the support reactions, then use the equilibrium of the joint with only two unknown member forces (joint B).

$\Sigma M_A = 0$: $6R_B = 24 \times 2$, so $R_B = 8$ kN ($R_A = 16$ kN).

BC has length $\sqrt{(4^2 + 3^2)} = 5$ m. Joint B, vertical: $8 + F_{BC} \times 3/5 = 0 \rightarrow F_{BC} = -13.33$ kN (compression).

Horizontal at B: $-F_{AB} - F_{BC} \times 4/5 = 0 \rightarrow F_{AB} = 13.33 \times 0.8 =$ **10.67 kN (tension)**

Exam tip: Assume every member in tension; a negative answer then means compression automatically.

Q3 **Answer: (C) 59.0°**

Concept: With a smooth wall, all the friction is at the floor; take moments about the foot.

Wall reaction R_w = friction $F = \mu W$ at limiting equilibrium. Moments about the foot: $W(L/2)\cos\theta = R_w L \sin\theta$

$\tan\theta = 1/(2\mu) = 1/0.6 \rightarrow \theta =$ **59.0° (C)**

Exam tip: A higher μ allows a flatter ladder; with friction at both surfaces use $\tan\theta = (1 - \mu_1\mu_2)/(2\mu_1)$.

Q4 **Answer: 23.69 (accept 23.50 to 23.85)**

Concept: Range on level ground $R = u^2 \sin 2\theta/g$; two complementary angles give the same range.

$\sin 2\theta = Rg/u^2 = 30 \times 9.81/400 = 0.7358$

$2\theta = 47.37° \rightarrow \theta =$ **23.69°** (the other angle is 66.31°)

Exam tip: The two angles are θ and $90° - \theta$; maximum range $u^2/g = 40.8$ m occurs at 45°.

Q5 **Answer: (B) 9.34 kW**

Concept: Useful power = rate of gain of potential energy = ρQgH; input = output/efficiency.

Output = $1000 \times (2/60) \times 9.81 \times 20 = 6540$ W

Input = $6540/0.7 = 9343$ W ≈ **9.34 kW (B)**

Exam tip: Divide by efficiency to get the input; multiplying (4.58 kW) is a common slip.

Q6 **Answer: 9**

Concept: Momentum is conserved; e fixes the separation speed: $v_2 - v_1 = e(u_1 - u_2)$.

Momentum: $1 \times 4 + 2 \times (-2) = 0 = v_1 + 2v_2$; restitution: $v_2 - v_1 = 0.5 \times (4 + 2) = 3$

$v_2 = 1$ m/s, $v_1 = -2$ m/s. KE before = ½ × 1 × 16 + ½ × 2 × 4 = 12 J; after = ½ × 1 × 4 + ½ × 2 × 1 = 3 J

Loss = **9 J** (in general $\Delta KE = \frac{1}{2} m_1m_2/(m_1 + m_2) \times (1 - e^2)(u_1 - u_2)^2 = \frac{1}{2} \times (2/3) \times 0.75 \times 36 = 9$ J ✓)

Exam tip: Velocities are vectors: take one direction positive and give the opposite velocity a minus sign.

Q7 **Answer: (A) 4/3**

Concept: $E = 2G(1 + \nu) = 3K(1 - 2\nu)$.

$2.4G = 2G(1 + \nu) \rightarrow \nu = 0.2$

$K = E/[3(1 - 2\nu)] = 2.4G/1.8 \rightarrow K/G =$ **4/3 (A)**

Exam tip: Find ν first; it links all the elastic constants.

Q8 **Answer: 0.398 (accept 0.395 to 0.401)**

Concept: For a uniformly tapering circular bar, $\delta = 4PL/(\pi E\, d_1\, d_2)$.

$\delta = 4 \times 50\,000 \times 1000/(\pi \times 200\,000 \times 40 \times 20)$

$\delta = 2 \times 10^8/(5.027 \times 10^8) =$ **0.398 mm**

Exam tip: Using the mean diameter (30 mm) gives 0.354 mm, which is wrong; the area varies as the square of the diameter.

Q9 **Answer: 80**

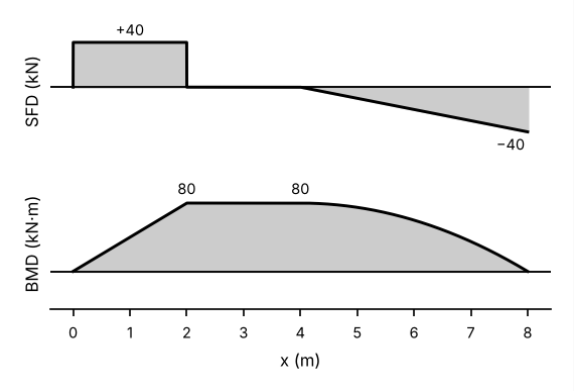
Concept: The maximum bending moment occurs where the shear force is zero or changes sign.

$\Sigma M_B = 0$: $8R_A = 40 \times 6 + (10 \times 4) \times 2 = 320 \rightarrow R_A = 40$ kN, $R_B = 80 - 40 = 40$ kN

Shear: +40 kN from A to the point load, then $40 - 40 = 0$ from 2 m to 4 m, so the BM is constant there.

$M = 40 \times 2 =$ **80 kN·m**, constant from x = 2 m to x = 4 m (SFD and BMD below).

Exam tip: A zone of zero shear means a flat portion of the BMD: pure bending between the two loads.



Q10 **Answer: (D) $bh^2/24$**

Concept: $Z = I/y_{max}$, with y measured from the neutral (centroidal) axis.

I about the centroidal axis = $bh^3/36$. The centroid is h/3 above the base, so the vertex is 2h/3 away.

$Z_{top} = (bh^3/36)/(2h/3) =$ **$bh^2/24$ (D)**; for the base fibre $Z = bh^2/12$.

Exam tip: In an unsymmetrical section the maximum bending stress is at the fibre farthest from the neutral axis (the vertex here).

Q11 **Answer: 5.63 (accept 5.58 to 5.67)**

Concept: Beyond the load the beam is unloaded, so it stays straight with the slope at C.

At C (a = 1.5 m): $y_C = Wa^3/(3EI) = 12 \times 3.375/18\,000 = 2.25$ mm; $\theta_C = Wa^2/(2EI) = 12 \times 2.25/12\,000 = 0.00225$ rad

$y_B = y_C + \theta_C \times 1.5 = 2.25 + 3.375 =$ **5.63 mm** $(= 5WL^3/(48EI))$

Exam tip: For a load at mid-length of a cantilever the tip deflection is 5/16 of that for the same load at the tip.

Q12 **Answer: (B) Tb/L**

Concept: Statically indeterminate: $T_A + T_B = T$ and the twist of AC equals the twist of CB.

$T_A\, a/(GJ) = T_B\, b/(GJ) \rightarrow T_A/T_B = b/a$

T_A = **Tb/L (B)**: the shorter (stiffer) part carries the larger torque.

Exam tip: Same as an axially loaded bar fixed at both ends: each reaction is proportional to the length of the other part.

Q13 **Answer: (C) 125 MPa**

Concept: The hoop stress acts across the longitudinal joint, so it is divided by the longitudinal joint efficiency.

$\sigma_h = pD/(2t\eta_l) = 2 \times 800/(2 \times 8 \times 0.8)$

σ_h = **125 MPa (C)**; the longitudinal stress uses η_c: $pD/(4t\eta_c) = 100$ MPa.

Exam tip: Longitudinal joint ↔ hoop stress; circumferential joint ↔ longitudinal stress. Swapping them is the usual error.

Q14 **Answer: (A) L/√2**

Concept: Effective length = distance between points of zero bending moment (inflection points) of the buckled shape.

Both ends hinged: L; both fixed: L/2; fixed–free: 2L; fixed–hinged: **$L/\sqrt{2} \approx 0.7L$ (A)**

The buckling load is then $P = 2\pi^2EI/L^2$, twice that of a pin-ended column.

Exam tip: $P \propto 1/Le^2$: the ratios for (fixed–free : pinned : fixed–pinned : fixed–fixed) are 1/4 : 1 : 2 : 4.

Q15 **Answer: 62.99 (accept 62.70 to 63.30)**

Concept: $\sigma n = (\sigma x + \sigma y)/2 + (\sigma x - \sigma y)/2 \cos 2\theta + \tau xy \sin 2\theta$, with θ measured counter-clockwise from x to the normal.

$= 40 + 20 \times \cos 60° + 15 \times \sin 60° = 40 + 10 + 12.99$

σn = **62.99 MPa**

Exam tip: Check with $\sigma_1 = 40 + \sqrt{(20^2 + 15^2)} = 65$ MPa: no normal stress can exceed it.

What next?

Day 69 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 70/90: TOM, Vibrations & MD Revision. Comment **GATE90** on our reel to get all links.
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